Exact checks of Semaev's S_3 and the summation-cover monodromy in characteristics 2 and 3

2026-10-29

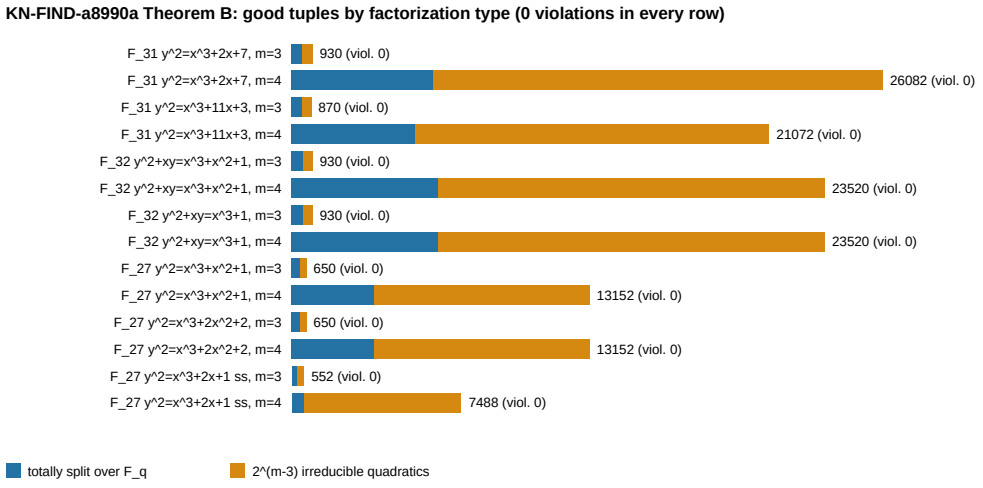
Why. The 2026-10-29 exposure audit (docs/reports/2026-10-29-cfsg-lessons-ecdlp-exposure/) found two summation-polynomial claims that rest on unchecked premises:

- Semaev's correspondence (KR-IC-5a43e5) is filed proven from abstract reads only. No Lean target or computation in this repository checks it against the group law.
- KN-FIND-a8990a states its monodromy and factorization theorems "in every characteristic incl. 2, 3". Its 241,643 verified specializations are all over prime fields.

What was run. python3 research/verification/semaev_char23_checks_20261029.py result.json (script, result.json), in under a minute on one CPU. It is an exact, exhaustive computation over small fields: no sampling, no statistics.

1. Semaev S_3 against the group law. The script takes every ordered pair of distinct abscissae x_1, x_2 in F_q. It lifts them to points P_1, P_2 over F_{q²}, and checks that the published $S_3(x_1, x_2, X)$ equals $lc \cdot (X - x(P_1+P_2)) \cdot (X - x(P_1-P_2))$ coefficient by coefficient. Two formulas are checked:
   - short Weierstrass in odd characteristic $p > 3$;
   - $y^2 + xy = x^3 + a_2x^2 + a_6$ in characteristic 2.
2. Theorems B and C of KN-FIND-a8990a at m = 3 and m = 4. The script uses generic long-Weierstrass arithmetic, so no S_m formula enters. For every good tuple a ∈ F_q^{m−1}, it builds the polynomial whose roots are the signed-sum abscissae. It checks three things:
   - the coefficients lie in F_q and the roots are distinct;
   - either every root is in F_q (exactly when the fibre classes c(a_i) agree) or none is;
   - the exact class count S^k N^{n−k} + N^k S^{n−k}.
3. Known-false controls on the same instrument. Two deliberately wrong inputs:
   - S_3 with its constant term shifted by one;
   - the fibre class of one rational abscissa flipped.

Result



factorization types

curve	q	S_3 pairs, mismatches	m = 3 good, violations	m = 4 good, violations	controls rejected
$y^2 = x^3+2x+7$	31	930, 0	930, 0	26,082, 0	S_3 930/930; class 60
$y^2 = x^3+11x+3$	31	930, 0	870, 0	21,072, 0	S_3 930/930; class 58
$y^2+xy = x^3+x^2+1$	32	992, 0	930, 0	23,520, 0	S_3 992/992; class 60
$y^2+xy = x^3+1$	32	992, 0	930, 0	23,520, 0	S_3 992/992; class 60
$y^2 = x^3+x^2+1$ (ordinary)	27	not checked	650, 0	13,152, 0	class 50
$y^2 = x^3+2x^2+2$ (ordinary)	27	not checked	650, 0	13,152, 0	class 50
$y^2 = x^3+2x+1$ (supersingular)	27	not checked	552, 0	7,488, 0	class 46

The claims hold, and the checker can tell when they don't.

- Every pair satisfies Semaev's S_3 correspondence in characteristic 31 and characteristic 2.
- Every good tuple obeys the totally-split-or-all-quadratics dichotomy at m = 3 and m = 4 in characteristics 31, 2 and 3. This includes a supersingular curve in characteristic 3, which KN-FIND-a8990a's "every curve" wording covers.
- The class-count law holds exactly in every case.
- Both known-false controls fail on every curve. The perturbed S_3 mismatches on every pair, and the flipped class map produces 46–60 violations.

Limits

- Small fields only. These are exact checks at $q \leq 32$ and $m \leq 4$. They corroborate:
  - Semaev's correspondence at m = 3;
  - KN-FIND-a8990a Theorems B and C at $m \leq 4$ in characteristics 2 and 3.

  They do not prove either claim in general, and they say nothing about ECDLP cost.
- No closed S_3 in characteristic 3. This repository has no published closed-form S_3 for characteristic 3. There, only the point-defined polynomial is checked, which is what Theorems B and C are about.
- No review yet. KN-FIND-a8990a's own review gap (AGENTS.md rule 12) is unchanged. A validator still has to read its derivation.
- Graphs checked. Only factorization.svg is new. No existing dashboard graph plots these quantities, so none needed updating.